

AGENEWTECH

H1504TQ

Smart Module Specification_V1.5



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About the Document

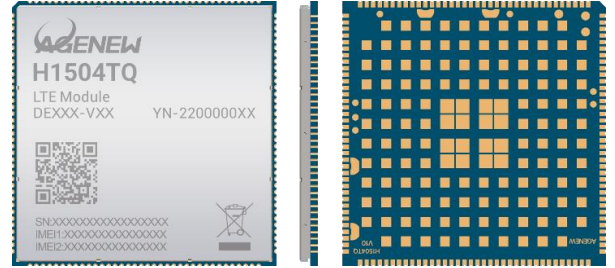
History

Version	Data	Author	Description
-	2025-07-28	Wu Danyi	Document creation
1.0	2025-07-30	Wu Danyi	Controlled Copy
1.1	2025-08-06	Wu Danyi	Modify the description of UART0 and UART1
1.2	2025-08-25	Wu Danyi	Modification area
1.3	2025-10-30	Wu Danyi	Update the Android version and memory type
1.4	2026-01-13	Wu Danyi	Add Latin American frequency band
1.5	2026-03-24	Chen Yun	Change company name

H1504TQ

Multi-network Smart 4G module

Supports Wi-Fi & BT



H1504TQ is a multi-network Smart LTE Cat 4 module based on MediaTek quad-core ARM Cortex-A53 processor (with a maximum clock speed of 1.3GHz) launched by Agenew, equipped with Android 10 or 13 operating system. It has powerful performance, neon engine processing, and 2D/3D graphics acceleration processing capabilities, fully meeting customers' needs for high speed and multimedia functionality in industrial and consumer applications.

H1504TQ supports full coverage of all network systems, WI-FI 802.11a/b/g/n, and BT4.2 near-field wireless communication. In addition, the H1504TQ also has audio and video input and output interfaces and rich GPIO interfaces. It also supports a variety of audio and video codecs. The main screen supports HD+(1440×720)@60fps at most. At the same time, H1504TQ supports GPS, GLONASS, BDS or Galileo positioning systems, making its positioning function fast and accurate.

H1504TQ integrates feature rich interfaces (such as LCM, camera, touch screen, microphone, speaker, UART, USB, I2C, SPI, SDIO, and ADC interfaces), greatly expanding its application in the M2M field, making it widely used in smart cash registers, smart POS, tax control machines, security monitoring, vehicle equipment, and high-end information Collection equipment, smart robots, smart homes, smart hardware, industrial smart handheld devices, drones, high-end police or law enforcement equipment, smart intercom devices, smart wearables, vending machines, logistics cabinets and other equipment are used in different industries.



Main Advantages

- ✓ Comprehensive coverage of multi-network standard
- ✓ Support Wi-Fi 802.11a/b/g/n and BT4.2 short-range wireless communication
- ✓ Support GPS/GLONASS/BDS/ Galileo
- ✓ Support multiple audio and video codecs
- ✓ The main screen supports up to HD+(1440×720)@60fps
- ✓ Support 1080P@30fps video recording and playback
- ✓ Possess audio and video input and output interfaces and rich GPIO interfaces



quad-core ARM Cortex-A53
64-bit processor



IEEE 802.11
a/b/g/n



Support Android 10
and Android 13



GPS+Glonass+BDS+Galileo



Powerful multimedia



BT4.2

LTE Cat4		H1504TQ	
Region/Operator	China/Eurasia/Latin America		
Operating System	Android 10 / Android 13		
Memory	8GB eMMC + 1GB LPDDR3 / 16GB eMMC + 2GB LPDDR3		
Module size (mm)	40.5mm(±0.1mm) × 40.5mm(±0.1mm) × 2.8mm(±0.2mm)		
Packaging	LCC + LGA		
Moisture Sensitivity Level	MSL3		
Temperature Range			
Operation Temperature	Normal operating temperature: -20 °C ~ + 80°C Storage temperature range: -40 °C ~ + 90 °C		
RF Band Information	P0	P1	
LTE-FDD	B1/B3/B5/B7/B8/B20/B28A/B28B	B1/B2/B3/B4/B5/B7/B8/B28A/B28B	
LTE-TDD	B38/B40/B41(194M)	B38/B40	
WCDMA	B1/B2/B5/B8	B1/B2/B5/B8	
GSM/EDGE	B2/B3/B5/B8	B2/B5	
WLAN	2.4/5GHz, 802.11 a/b/g/n		
Bluetooth	4.2		
GNSS	GPS/GPS+GLONASS/GPS+BDS		
Authentication			
Operator Certification	Undetermined		
Compulsory Certification	SRRC*/NAL*/CCC*		
Other Certifications	RoHS*/REACH*/CE*		
Data Rate			
LTE-FDD (Mbps)	150Mbps (DL)/50Mbps (UL)		
LTE-TDD (Mbps)	130Mbps (DL)/30Mbps (UL)		
HSPA (Mbps)	21Mbps (DL)/11Mbps (UL)		
WCDMA (Kbps)	384Kbps (DL)/384Kbps (UL)		
EDGE (Kbps)	296Kbps (DL)/236.8Kbps (UL)		
GPRS (Kbps)	107Kbps (DL)/85.6Kbps (UL)		
Outstanding Features			
3GPP-E-UTRA	LTE: Release 10 UMTS: Release 9		
Bandwidth（MHz）	1.4-20 MHz		
SIM Card Detection	●		
Dual SIM Dual Standby	●		
USB Firmware Upgrade	●		
OTA Firmware Upgrade	●		

Remarks: 1. *: In progress

2. ●: Supports

3. Extended operating temperature: Working within this temperature range may result in a decrease in RF performance, but it can still function normally.

LTE Cat4	H1504TQ
Electrical specification	
Operating Voltage	3.5~4.4 V, typical 3.8 V
Consumption (typical)	0.35mA@Shutdown mode; typical 3mA@Sleep mode
Interface	
LCD	4-lane MIPI_DSI, Resolution up to HD+1440×720@60fps
Camera	Front camera 1-lane MIPI_CSI, supporting up to 2MP photography Rear camera 4-lane MIPI_CSI, supporting up to 13MP photography
TP	Capacitive touch screen, I2C control
Audio	3 analog inputs: MIC1, MIC2, HEADMIC (including 1 noise cancelling microphone) 3 analog outputs: speaker, earpiece, headphone
USB	×1, USB 2.0 specification, Supports OTG, charge
(U)SIM	×2, supports 1.8/2.95 V (U) SIM cards, supports (U)SIM card hot plug detection, dual card dual standby
UART	×2, UART1 supports hardware flow control; UART0 is only used for debugging
SD Card	×1, SD3.0, 4-bit SDIO
ADC	×1
I2C	×4
SPI	×3, Reusable as GPIO
GPIO	×25
Motor driven	×1
PWRKEY	Internal pull up to VBAT
ANT	Main antenna, Diversity antenna, GNSS antenna, WIFI/BT antenna